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Case 22-2021: A 64-Year-Old Woman with Cognitive Impairment, Headache, and Memory Loss

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PRESENTATION OF CASE

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Dr. Giovanna S. Manzano (Neurology): A 64-year-old woman was admitted to this hospital with cognitive impairment, headache, and memory loss.

The patient had been well until 6 weeks before admission, when respiratory symptoms developed. Testing of a nasopharyngeal swab for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) RNA was negative. At around the same time, coworkers observed a decline in the patient's performance at work. The patient noticed that she was having difficulty operating computers and was forgetting her personal passwords.

The patient was evaluated at another hospital. Although she reported recent confusion and changes in cognitive performance, the examination at that time — including a neurologic examination with assessment of orientation, speech, and memory recall after 5 and 20 minutes — was unremarkable. The patient was referred for an outpatient neurologic evaluation.

One week later (5 weeks before admission), severe headache developed suddenly, along with blurred vision, dizziness, and nausea. After the administration of ondansetron, the blurred vision, dizziness, and nausea resolved; after the administration of nonsteroidal antiinflammatory drugs, the headache became less severe, with mild pain and intermittent episodes of more severe pain.

Three weeks before admission, the patient was seen in a neurology clinic affiliated with the other hospital. She had difficulty recalling the history of her symptoms. On examination, she was oriented to place and person only and was unable to perform serial sevens calculations or to spell “world” backward. The remainder of the examination was unremarkable. The complete blood count with differential count and test results for liver and kidney function were normal, as were blood levels of electrolytes, glucose, thiamine, and cobalamin. Tests for syphilis and human immunodeficiency virus infection were negative; other labora-

tory test results are shown in Table 1. An electroencephalogram (EEG) showed no distinct epileptiform activity. Topiramate was prescribed for a working diagnosis of complex migraine. Imaging studies were obtained.

Dr. R. Gilberto Gonzalez: Magnetic resonance imaging (MRI) of the head was performed. T2-weighted fluid-attenuated inversion recovery (FLAIR) images showed diffusely scattered hyperintensities in the periventricular, subcortical, and deep white matter and leptomeningeal enhancement. In addition, there was evidence of frontal superficial siderosis and a few scattered microhemorrhages.

Dr. Manzano: Lumbar puncture was also performed. On cerebrospinal fluid (CSF) analysis, the white-cell count was 46 per microliter (reference range, 0 to 5), with a lymphocyte count of 99% (reference range, 40 to 80), the red-cell count was 35 per microliter (reference value, 0), and the protein level was 221 mg per deciliter (reference range, 13 to 40); other laboratory test results are shown in Table 1. Cultures of the CSF were negative. The administration of topiramate was stopped; treatment with oral doxycycline was started but then was stopped after 5 days, when results of a test for *Borrelia burgdorferi* were interpreted as negative.

Over the next 3 weeks, intermittent headache persisted and cognitive impairment worsened. The patient presented to the emergency department of this hospital. She was unable to provide a history. According to the patient's husband, the patient had stopped working and was no longer able to effectively care for herself, requiring assistance with daily activities such as bathing, dressing, and eating. She had difficulty planning and was unable to put on clothes in the correct sequence or to turn on the shower. Memory impairment persisted, conversation was tangential, and visual hallucinations frequently occurred. The patient had lost 6.8 kg in the past month. There was no fever, rash, or joint pain.

The patient had a history of poliomyelitis in childhood, with residual mild weakness in the left leg; vitamin B₁₂ deficiency, which had been corrected; and hypothyroidism. Medications included atorvastatin, levothyroxine, sertraline, vitamin B₁₂, and cholecalciferol. She had a known allergy to amoxicillin-clavulanate with an unknown reaction. She was a former smoker with

a 40 pack-year history, and she had consumed four alcoholic beverages per week before the onset of the current illness. The patient worked in a health care clinic and lived with her husband in a suburban area of New England. She had not traveled outside the United States. She was an avid gardener; she had a dog but no other animal exposures. Her father had received a diagnosis of Alzheimer's disease at 62 years of age.

On examination, the patient was alert and interactive. The temperature was 36.7°C, the blood pressure 105/51 mm Hg, the heart rate 56 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 96% while she was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 32.4. The patient was oriented to time, place, and person. Her speech was fluent, with intact naming, repetition, and comprehension. She had impaired abstraction and mildly impaired attention. She was unable to recall three objects after a delay. Intermittently during the examination, she had visual hallucinations that she knew were not real. Cranial nerve examination was unremarkable. Strength was 5/5 throughout, except in the left leg (4/5). Reflexes were brisk throughout, including 3+ reflexes in the arms and the right lower leg; pectoralis reflexes and a jaw jerk were present. No ankle clonus was noted. Examinations for tone, sensation, and cerebellar function were unremarkable. The remainder of the examination was normal. Laboratory test results are shown in Table 1.

Dr. Gonzalez: MRI of the head was performed (Fig. 1). FLAIR images showed slight progression of the diffuse subarachnoid hyperintensities and leptomeningeal enhancement. Susceptibility-weighted images showed frontal superficial siderosis and scattered lobar microhemorrhages. Computed tomographic (CT) angiography of the head revealed patent cranial vessels, without evidence of aneurysm, arteriovenous malformation, or thrombosis. A CT scan of the chest and abdomen, obtained after the administration of intravenous contrast material, was normal.

Dr. Manzano: Lumbar puncture was performed. On CSF analysis, the nucleated-cell count was 28 per microliter (reference range, 0 to 5), the lymphocyte count 85%, the red-cell count 53 per

Table 1. Laboratory Data.*

Variable	Reference Range, Other Hospital	3 Wk before Admission, Other Hospital	Reference Range, This Hospital†	On Admission, This Hospital
Blood				
Thyrotropin (μ IU/ml)	—	—	0.40 to 5.00	8.65
Free thyroxine (ng/dl)	—	—	0.9 to 1.8	1.4
Antithyroid peroxidase antibodies (IU/ml)	—	—	<35.0	32.4
IgG (mg/dl)	586 to 1602	946	767 to 1590	814
Albumin (g/dl)	3.8 to 4.8	5.2	3.2 to 4.8	4.24
Interferon-gamma release assay	Negative	Negative	—	—
West Nile virus IgG and IgM	Negative	Negative	—	—
Q fever antibodies	Negative	Negative	—	—
Lyme disease antibodies	Negative	Negative	Negative	Negative
Brucella IgG and IgM	—	—	Negative	Negative
Anaplasma, PCR assay	—	—	Negative	Negative
Babesia, thick and thin smear	—	—	Negative	Negative
Urine				
Histoplasma antigen	—	—	Negative	Negative
Cerebrospinal fluid				
Red-cell count (per μ l)	0	35	0 to 5	53
White-cell count (per μ l)	0 to 5	46	0 to 5	28
Differential count (%)				
Lymphocytes	40 to 80	99	0 to 100	85
Neutrophils	0 to 7	1	0	1
Monocytes	15 to 45	0	0 to 100	14
Protein (mg/dl)	13 to 40	221	5 to 55	280
Glucose (mg/dl)‡	30 to 70	58	50 to 75	58
Gram's stain	—	No bacteria seen	—	No bacteria seen
Xanthochromia	—	—	—	Present
Oligoclonal bands	—	Not seen	—	Not seen
IgG (mg/dl)	0 to 8.6	28.1	$\leq$ 8.1	34.6
IgG index	0 to 0.7	5.2	$\leq$ 0.85	1.05
IgG synthesis rate	−9.9 to 3.3 per day	75.9	$\leq$ 12 per day	100.43
Albumin (mg/dl)	11 to 48	155	$\leq$ 27.0	172.0
Albumin index	0 to 8	30	$\leq$ 0.40	0.19
Herpes simplex viruses 1 and 2 DNA	Negative	Negative	Negative	Negative
West Nile virus RNA	—	—	Negative	Negative
Varicella–zoster virus, PCR assay	—	—	Negative	Negative
VDRL test	—	—	Negative	Negative
Lyme disease				
IgG	—	p93, p28, p23	—	—
IgM	—	p39	—	—
Listeria antibodies	—	—	<1:1	<1:1

Table 1. (Continued.)

Variable	Reference Range, Other Hospital	3 Wk before Admission, Other Hospital	Reference Range, This Hospital†	On Admission, This Hospital
Histoplasma antibodies	—	—	Negative	Negative
Enterovirus RNA	—	—	Negative	Negative
Epstein-Barr virus DNA	—	—	Negative	Negative
Cryptococcal antigen	—	—	Negative	Negative
Arbovirus antibodies	—	—	Negative	Negative

* PCR denotes polymerase chain reaction, and VDRL Venereal Disease Research Laboratory.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

‡ To convert the values for glucose to millimoles per liter, multiply by 0.05551.

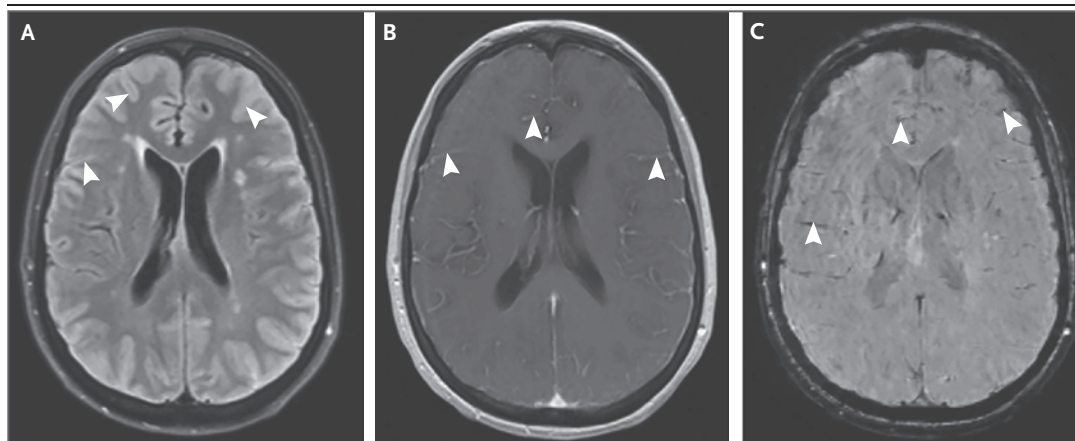


Figure 1. MRI of the Head.

A T2-weighted fluid-attenuated inversion recovery image (Panel A) shows abnormal diffuse sulcal hyperintensity (arrowheads). A T1-weighted image obtained after the administration of intravenous gadolinium (Panel B) shows abnormal diffuse leptomeningeal enhancement (arrowheads). A susceptibility-weighted image (Panel C) shows abnormal sulcal hypointensity in multiple locations (arrowheads).

microliter, and the protein level 280 mg per deciliter. CSF was sent for culture and cytologic studies; other laboratory test results are shown in Table 1. An EEG was normal.

The patient was admitted to this hospital, and intravenous ceftriaxone, acyclovir, and methylprednisolone were administered. During the first 48 hours, the patient was agitated and impulsive. Visual hallucinations worsened, and olanzapine was administered. The CSF culture was negative, and the CSF cytologic studies showed no malignant cells.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Shamik Bhattacharyya: When approaching the differential diagnosis in a complex case such as this one, I start by naming the clinical syndrome as a broad category and then consider the potential underlying causes.

CLINICAL SYNDROME

This patient's initial neurologic symptoms were cognitive. She had declining performance at work and forgot computer passwords. These mild symptoms progressed to disorientation and

inattention, as indicated by the inability to perform serial sevens calculations. This cognitive deterioration can be broadly described with the term “encephalopathy,” which implies brain parenchymal dysfunction. Selective networks of cognition could have been affected, with specific neuroanatomical localization; for example, frontal executive dysfunction could have caused difficulty operating a computer. However, we do not have results from a detailed neurocognitive examination obtained early in this patient’s disease course, so we cannot be more specific.

The patient’s second set of clinical symptoms—including severe headache, nausea, and blurred vision—is associated with meningeal irritation and improved with treatment. Finally, the patient had evidence of inflammation, including the subacute evolution of the syndrome over a period of weeks, the presence of pleocytosis on CSF analysis, and the presence of gadolinium enhancement on MRI of the head. If we tie together the findings of encephalopathy, meningeal irritation, and inflammation, the clinical syndrome is consistent with meningoencephalitis.

MENINGOENCEPHALITIS

Finding a specific cause of meningoencephalitis can be a difficult clinical task, often requiring the integration of history, risk factors, and results from multiple testing techniques, rather than reliance on a single testing strategy. No single test or panel of tests is comprehensive enough to identify all or even most causes of encephalitis. For example, in a prospective series involving patients with meningitis and encephalitis who were evaluated clinically and with metagenomic sequencing of CSF for infectious diseases, approximately 50% of cases had no specific cause identified.¹ This finding parallels results of earlier studies, in which a substantial fraction of encephalitis cases did not have a clearly identifiable cause.^{2,3} When I consider causes of meningoencephalitis, I first divide them into infectious and noninfectious causes.

Infectious Causes

The list of infections that could potentially lead to encephalitis is long and can be narrowed in this case by considering the patient’s risk factors and exposure history. The patient was not known to be immunocompromised, and she did

not have a history of travel outside the northeastern United States. The most commonly identified cause of acute encephalitis in older adults is herpes simplex virus encephalitis, which typically involves the temporal lobes.^{4,5} Features in this patient that argue against this diagnosis are the relatively slow pace of evolution (over a period of weeks, rather than days), the absence of fever, and the absence of temporal lobe involvement on MRI.

The most common type of arboviral meningoencephalitis in the northeastern United States is West Nile virus encephalitis, with rarer types including Powassan virus encephalitis and eastern equine encephalitis.⁶ The time course of this patient’s illness is more indolent than would be expected with these infections. Characteristic imaging features of arboviral encephalitis, including involvement of the deep gray matter, cerebellum, and brain stem, were absent in this case.⁷ Varicella–zoster virus, which notoriously causes variable forms of encephalitis, remains a possibility.⁸

This patient had disparate results of Lyme disease testing, with a positive CSF test showing IgG and IgM bands but with a negative blood test. In testing for Lyme disease, a serologic test of the blood for antibodies to *B. burgdorferi* is performed first, and then a CSF test is performed if indicated on the basis of signs and symptoms.⁹ This patient had no detectable Lyme disease antibodies in the blood, and therefore, the relevance of positive bands on CSF testing is unclear. Although meningoencephalitis can be caused by Lyme disease, this patient did not have rash, arthritis, or radiculitis, features that would further suggest the diagnosis of Lyme disease. In addition, false positive results of CSF serologic testing for Lyme disease in patients without any signs of neuroborreliosis can occur.¹⁰

Tuberculosis can cause subacute meningoencephalitis. However, this infection is not common in the northeastern United States, and the absence of fever and the negative interferon-gamma release assay make it an unlikely diagnosis in this case.¹¹ Cryptococcal infection can cause subacute meningoencephalitis in an immunocompetent person. However, the CSF cryptococcal antigen test is highly sensitive for cryptococcal disease and was negative in this patient.¹²

Noninfectious Causes

Immune-mediated noninfectious causes of meningoencephalitis can result in subacute neurologic syndromes that affect both the central and peripheral nervous systems. The rapidly emerging knowledge about these syndromes can be overwhelming, given that new disease-causing antineural antibodies targeting obscure antigens are discovered and reported frequently. As with infectious causes of meningoencephalitis, a systematic approach to noninfectious causes, focusing on risk factors, may help in finding a cause and proposing a treatment.

In the United States, demyelinating diseases are the most common autoimmune neurologic disorders of the brain, and they should be considered in this case. Multiple sclerosis is unlikely in this patient because of the absence of typical demyelinating lesions on MRI of the head, the absence of oligoclonal bands in the CSF, and the presence of headache, which is an atypical symptom in patients with multiple sclerosis.^{13,14} Among related demyelinating disorders, acute disseminated encephalomyelitis and anti-myelin oligodendrocyte glycoprotein syndrome can cause subacute meningoencephalitis, but MRI typically shows multifocal cortical and subcortical hyperintense lesions on T2-weighted imaging, which were not present in this patient.¹⁵

Neurosarcoidosis is less rare than previously thought, and in approximately 50% of cases, neurologic symptoms are the first manifestation.¹⁶ Subclinical evidence of extraneural sarcoidosis, such as mediastinal lymphadenopathy, is often present on further investigation, even in cases in which neurologic symptoms predominate. A potentially helpful clue on MRI is the presence of tortuous deep medullary veins on gadolinium-enhanced and susceptibility-weighted imaging.¹⁷ Although CT of the chest and abdomen in this patient did not show enlarged lymph nodes and MRI of the head did not show tortuous deep medullary veins, neurosarcoidosis remains a possible diagnosis in this patient.

The other causes of noninfectious meningoencephalitis can be broadly grouped under the rubric of autoimmune encephalitis. Consensus clinical criteria for autoimmune encephalitis have been proposed, and this patient meets criteria for possible autoimmune encephalitis on the basis of progressive neuropsychiatric changes

over a period of less than 3 months, CSF pleocytosis, and the absence of a clear alternative explanation.¹⁸ When considering possible causes of autoimmune encephalitis, I pose the following seven questions.

First, is the patient's syndrome consistent with limbic encephalitis? Inflammation in the limbic regions of the brain causes subacute deficits in memory, personality changes, and seizures. Encephalitis associated with antibodies to leucine-rich glioma-inactivated 1 is a common type of limbic encephalitis in older adults, and hyponatremia may be a diagnostic clue.¹⁹ In this patient, the abnormalities on MRI of the head were present in meningeal and extratemporal regions, rather than in limbic regions such as the medial temporal lobes, as would be expected with limbic encephalitis.¹⁸

Second, is the syndrome consistent with encephalitis associated with antibodies to N-methyl-D-aspartate receptors? This is one of the most common causes of autoimmune encephalitis in younger patients, but it is an important diagnosis to consider in all age groups.²⁰ However, this patient did not have the characteristic progression of neuropsychiatric changes to seizures with speech disturbances, a movement disorder, decreased arousal, and dysautonomia.²¹

Third, does the patient have cancer or systemic autoimmune disease? Cancer can directly invade the nervous system or cause paraneoplastic inflammatory injury. This patient did not have a known cancer, either noted in her history or discovered during the current evaluation. She had hypothyroidism, which is often due to immune-mediated injury to the thyroid gland. Hashimoto encephalopathy, also known as steroid-responsive encephalopathy associated with autoimmune thyroiditis, is a form of encephalitis associated with the presence of antithyroid peroxidase or antithyroglobulin antibodies. The clinical presentation includes progressive confusion and decreased consciousness, and results of MRI of the head are usually normal or nonspecific.²² Although this patient's clinical course could be consistent with Hashimoto encephalopathy, testing for antithyroid peroxidase antibodies was negative, and the abnormal gadolinium enhancement on MRI of the head points away from this diagnosis.

Fourth, is the syndrome consistent with

Creutzfeldt–Jakob disease? Rapidly progressive cognitive impairment in an older adult is suggestive of this prion-mediated neurodegenerative disease. A movement disorder, especially myoclonus, is common and was not present in this patient. In addition, CSF pleocytosis and the absence of typical restricted diffusion in the cortex and striatal regions on MRI argue against a diagnosis of Creutzfeldt–Jakob disease.²³

Fifth, is the patient's syndrome consistent with primary central nervous system (CNS) vasculitis? Although this diagnosis is often invoked, primary CNS vasculitis is a rare disease, with an estimated incidence of approximately 2 cases per 1 million person-years.²⁴ The disease is difficult to diagnose; there is no definitive combination of history, examination, imaging, and laboratory features specific enough to establish the diagnosis without histopathological proof. Patients with primary CNS vasculitis can present with headache, neuropsychiatric changes, and meningeal gadolinium enhancement on MRI, so this possibility should remain on the differential diagnosis in this case.

Sixth, is the syndrome caused by cerebral amyloid angiopathy (CAA)–related inflammation? In an adult 40 years of age or older, this condition usually manifests as a subacute syndrome with headache, progressive behavioral changes, and a decrease in consciousness.²⁵ MRI usually shows lesions involving the subcortical white matter, as well as hemorrhagic lesions (cerebral hemorrhage, cerebral microbleeds, or cortical superficial siderosis). In this patient, MRI revealed multiple microhemorrhages in a lobar distribution, cortical siderosis, and subtle cortical hyperintensities on T2-weighted imaging, features that are all consistent with the diagnosis of CAA-related inflammation.

Finally, does the patient have CNS lymphoma? Lymphoma can mimic multiple diseases and can affect both the meninges and the brain parenchyma. This patient primarily had evidence of leptomeningeal disease on imaging, without parenchymal masses. Primary leptomeningeal lymphoma occurs infrequently, accounting for less than 10% of primary CNS lymphomas.²⁶ Furthermore, approximately two thirds of patients with primary leptomeningeal lymphoma have malignant cells on CSF cytologic studies, which were not identified in this patient.

No single clue from the history, imaging studies, or laboratory test results in this case is specific for a diagnosis, and I suspect that a brain and meningeal biopsy was ultimately required to establish a pathological diagnosis and direct treatment. Reasonable diagnostic possibilities include varicella–zoster virus meningoencephalitis, neurosarcoidosis, primary CNS vasculitis, and CAA-related inflammation. Because of the presence of multiple hemorrhagic lesions in the brain on MRI, I think that CAA-related inflammation is the most likely diagnosis in this case.

DR. SHAMIK BHATTACHARYYA'S DIAGNOSIS

Cerebral amyloid angiopathy–related inflammation.

PATHOLOGICAL DISCUSSION

Dr. Samantha N. Champion: A surgical biopsy of the right frontal cortex was performed. Examination of the biopsy specimen revealed sections of cerebral cortex with moderate subpial gliosis and scattered acellular granular basophilic aggregates suggestive of amyloid plaques (Fig. 2A). The overlying leptomeninges showed acute hemorrhage with focal deposition of fibrin; an area of cellular arachnoid contained medium-sized vessels with thickened eosinophilic walls (some with splitting), chronic inflammatory cells, and hemosiderin deposition (Fig. 2A and 2B).

Immunohistochemical staining for amyloid-beta protein showed a moderate burden of amyloid plaques, rare neuritic plaques, rare amyloid deposition in small parenchymal vessels, and deposition in several leptomeningeal vessels with associated inflammation (Fig. 2C). Immunohistochemical stains for inflammatory markers showed that the inflammatory component was predominantly composed of CD3+ T cells (Fig. 2D) with scattered CD20+ B cells. An immunohistochemical stain for varicella–zoster virus was negative.

CAA-related inflammation is an autoimmune reaction to amyloid-beta deposits in the cerebral vasculature.²⁷ Diagnostic criteria include evidence of at least one CAA-positive vessel that shows lymphocytes in the vessel wall or perivas-

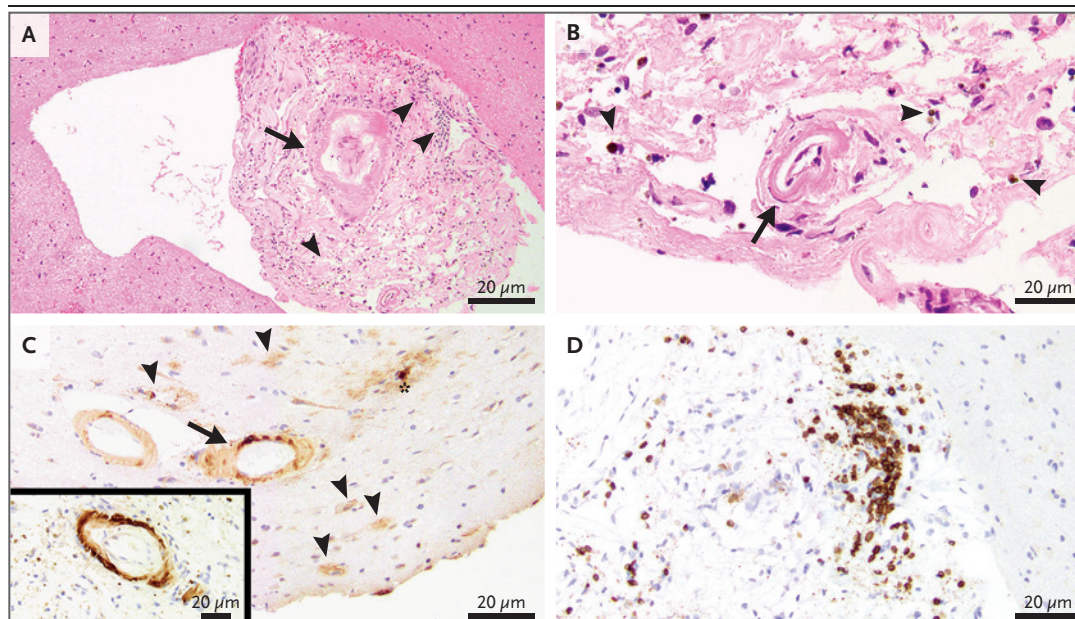


Figure 2. Biopsy Specimens of the Right Frontal Cortex.

On hematoxylin and eosin staining of sections of formalin-fixed and paraffin-embedded tissue (Panels A and B), the cerebral cortex shows moderate subpial gliosis and the overlying leptomeninges show medium-sized vessels with thickened eosinophilic walls (Panel A, arrow) and surrounding chronic inflammation (Panel A, arrowheads), as well as areas with vascular wall splitting (Panel B, arrow) and perivascular hemosiderin deposition (Panel B, arrowheads). Immunohistochemical staining for amyloid-beta (Panel C) shows a moderate amount of diffuse amyloid plaques (arrowheads), rare mature neuritic plaques (asterisk), rare amyloid deposition in small parenchymal vessels (arrow), and deposition in several leptomeningeal vessels (inset). Immunohistochemical staining for inflammatory markers (Panel D) shows that the leptomeningeal inflammation is predominantly composed of CD3+ T cells.

cular histiocytes without associated hemosiderin, since bland perivascular lymphocytes or perivascular macrophages with associated hemosiderin can be present with sporadic CAA.²⁵ In this case, there was no evidence of vessel-wall destruction or multinucleated giant cells, findings that are normally present with amyloid-beta-related angiitis.²⁸ Therefore, given the presence of leptomeningeal CAA and a prominent leptomeningeal perivascular inflammatory component without evidence of a destructive vasculitis or multinucleated giant cells, the features are most consistent with the diagnosis of CAA-related inflammation.

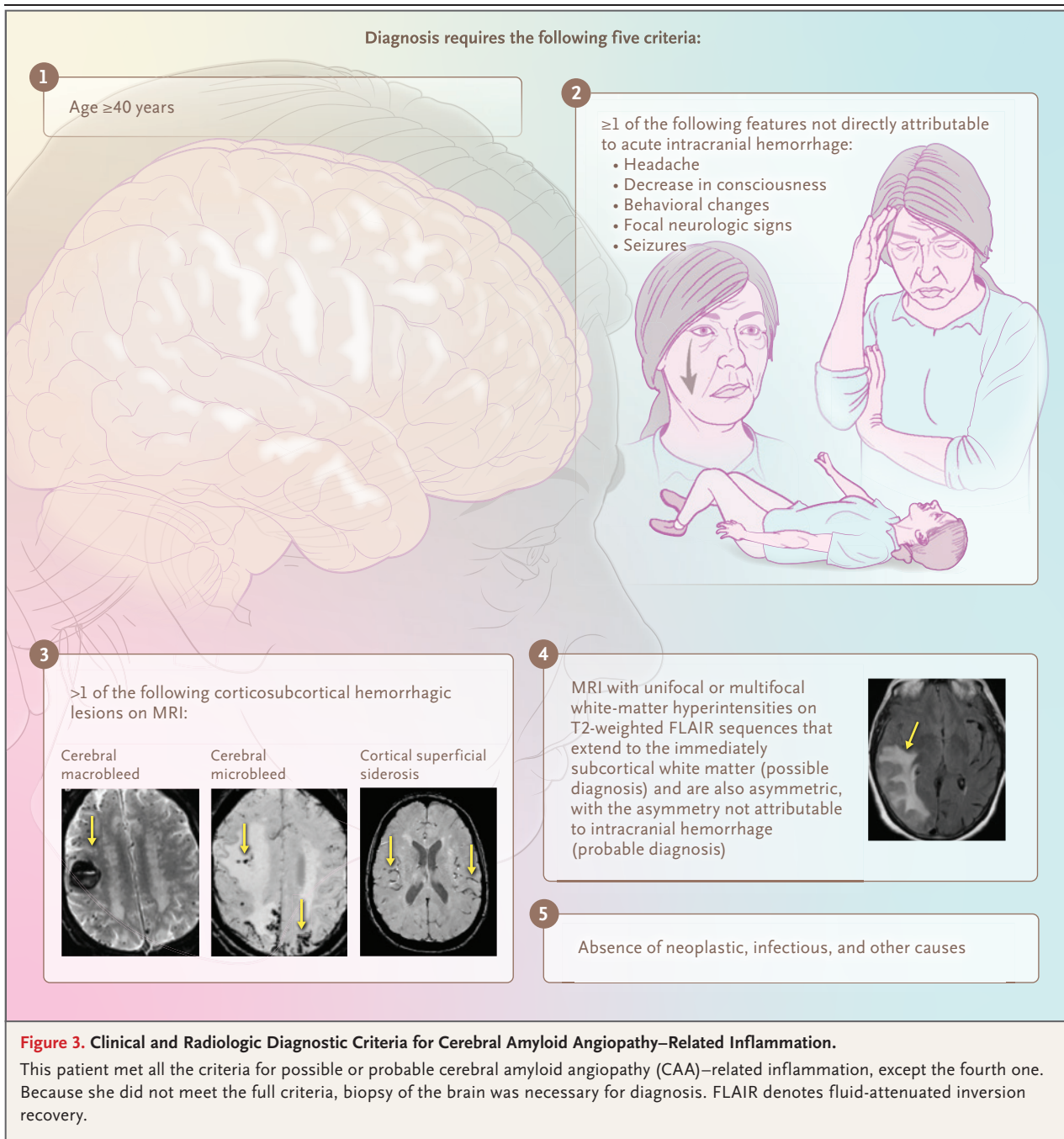
PATHOLOGICAL DIAGNOSIS

Cerebral amyloid angiopathy-related inflammation.

DISCUSSION OF MANAGEMENT

Dr. Bart K. Chwalisz: Most cases of inflammatory CAA have striking and characteristic abnormalities on MRI of the head, including unifocal or multifocal white-matter lesions that are bright on T2-weighted FLAIR sequences and may have mass effect, as well as cortical or subcortical hemorrhagic lesions that may include cerebral microbleeds or macrobleeds or cortical superficial siderosis. Published criteria based on these features have essentially made inflammatory CAA a clinoradiologic diagnosis (Fig. 3).²⁵ In this patient, although supportive hemorrhagic lesions were present, the typical bright lesions on T2-weighted imaging were absent, and thus a biopsy was necessary to confirm the diagnosis and rule out diseases that mimic CAA.

From the pathological perspective, inflam-



matory CAA includes two variants: amyloid-beta-related angiitis (which has more angiodestructive features)²⁹⁻³¹ and CAA-related inflammation.^{32,33} However, clinicopathological correlation can be imperfect, and thus treatment decisions for this patient were based on clinical

severity rather than on the pathological picture. A retrospective series involving 48 patients with inflammatory CAA showed clinicoradiologic improvement and a decreased risk of recurrence with immunosuppressive therapy.²⁷ Glucocorticoids are typically administered as first-line

therapy. Some patients have a monophasic disease course, but others may require additional glucocorticoid-sparing immunosuppressive medications; this approach is based on retrospective case reports and extrapolation from experience with primary angiitis of the CNS. In our practice, cyclophosphamide and mycophenolate mofetil are used in patients who are dependent on glucocorticoids or have a condition that is refractory to glucocorticoid treatment.

This patient was treated with intravenous methylprednisolone followed by a tapered course of oral prednisone. Despite some improvement in cognition and memory, her mental state remained impaired, and she continued to have visual hallucinations. Induction therapy with intravenous cyclophosphamide was commenced and resulted in somewhat improved alertness and orientation at the time of discharge from the hospital, although she continued to have limited insight regarding her visual hallucinations.

Dr. Manzano: The patient was seen in the neurology clinic of this hospital 1 month after discharge. She was living at home but required support from her husband for activities of daily living. She still had intermittent confusion and occasional hallucinations. She had

improved speech fluency and insight regarding her disease.

The patient was seen again in the neurology clinic of this hospital after completing three monthly doses of cyclophosphamide and a slowly tapered course of prednisone. She could independently perform activities of daily living, and her visual hallucinations had completely resolved. On examination, her attention was intact, her speech was fluent, and she was able to perform simple and two-step tasks, although she continued to have difficulty with recall testing.

Dr. Gonzalez: Repeat MRI of the head showed gradual resolution of the subarachnoid hyperintensities on T2-weighted FLAIR imaging. The chronic hemorrhagic changes persisted.

FINAL DIAGNOSIS

Cerebral amyloid angiopathy-related inflammation.

This case was presented at Neurology Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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